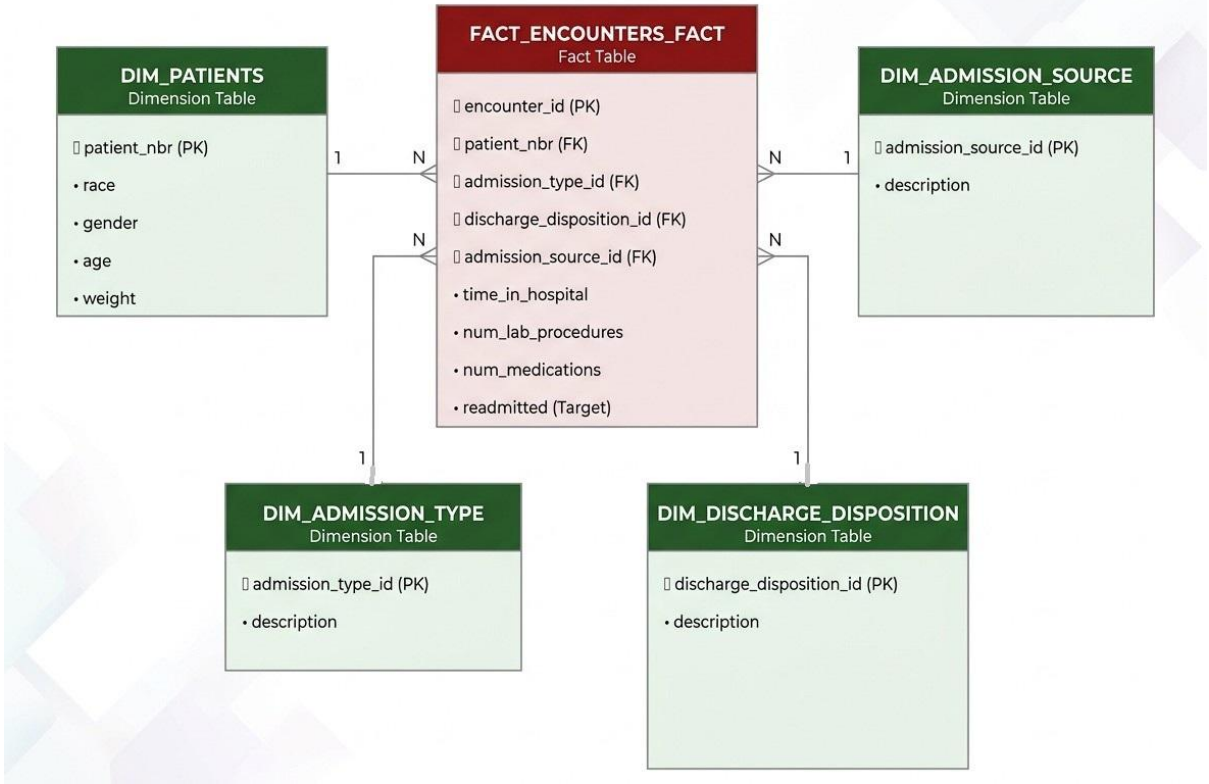
Data Model


FACT_ENCOUNTERS_FACT
Fact Table
encounter_id (PK)
patient_nbr (FK)
admission_type_id (FK)
discharge_disposition_id (FK)
admission_source_id (FK)
time_in_hospital
num_lab_procedures
num_medications
readmitted (Target)
DIM_PATIENTS
Dimension Table
patient_nbr (PK)
race
gender
age
weight
DIM_ADMISSION_SOURCE
Dimension Table
admission_source_id (PK)
description
DIM_ADMISSION_TYPE
Dimension Table
admission_type_id (PK)
description
DIM_DISCHARGE_DISPOSITION
Dimension Table
discharge_disposition_id (PK)
description
1
N
N
1
N
1
N
1

Data Cleaning Documentation

Diabetes Patient Dataset

1. Project Overview

This document describes the data cleaning steps performed on the diabetes patient dataset before starting the exploratory data analysis (EDA) stage.

2. Dataset Information

The dataset contains patient encounter information, demographic data, hospital visit details, diagnoses, laboratory results, medications, and readmission information.

3. Missing Values Cleaning

Missing values were checked and replaced with meaningful labels for categorical columns. The following columns were handled:

Column	Replacement
weight	Unknown
max_glu_serum	Not Tested
A1Cresult	Not Tested
medical_specialty	Unknown
payer_code	Unknown
race	Unknown
diag_1	Unknown
diag_2	Unknown
diag_3	Unknown

After cleaning, the dataset was checked again and no missing values remained.

4. Cleaning Gender Values

The value 'Unknown/Invalid' in the gender column was replaced with 'Unknown' to keep the category consistent.

5. Removing Constant Columns

The columns 'citoglipton' and 'examide' were checked and both contained only the value 'No' for all 101,766 records. Since they do not provide any variation or useful information for analysis, they were removed from the dataset.

6. Checking Patient and Encounter IDs

The dataset was checked for duplicate identifiers. There were 71,518 unique patient IDs and 30,248 repeated patient IDs. Repeated patient IDs were kept because the same patient can have multiple hospital encounters. The encounter_id column had no duplicates.

7. Creating Age as a Numeric Value

The original age column contains age ranges such as [70-80). A new column called age_start was created to extract the starting age from each range as a numeric value.

Example: [70-80) → age_start = 70

8. Creating Total Previous Visits

A new column called total_previous_visits was created by adding the previous outpatient, emergency, and inpatient visits:

total_previous_visits = number_outpatient + number_emergency + number_inpatient

9. Checking Numerical Values

The main numerical columns were reviewed using descriptive statistics. The minimum values were also checked to make sure there were no negative values.

No negative values were found in the checked numerical columns.

10. Checking Missing Values After Cleaning

The dataset was checked again using isnull().sum(). All checked columns had zero missing values.

11. Checking Duplicate Rows

The dataset was checked for complete duplicate rows. The result was 0 duplicate rows, so no rows needed to be removed.

12. Checking Data Types

The data types of all columns were reviewed. Numerical variables were stored as integer values, while categorical and text-based variables remained as object values. The original age column was kept as an object because it contains age ranges, while age_start was created as a numeric column.

13. Checking Categorical Values

Important categorical columns were reviewed to make sure their values were consistent and expected. The following columns were checked:

- change
- diabetesMed
- readmitted
- max_glu_serum
- A1Cresult

No unexpected categorical values were found, so no additional changes were required.

14. Final Dataset

After completing the cleaning process, the cleaned dataset was saved as:

diabetes_cleaned.csv

15. Data Cleaning Result

The dataset is now ready for the next stage of the project, which is Exploratory Data Analysis (EDA). The cleaning process focused on handling missing values, removing unnecessary constant columns, creating useful analytical features, checking data quality, and keeping valid repeated patient encounters.